

X3 / MAINNET-READINESS AUDIT ARTIFACT

X3 ATOMIC STAR

THE ROAD TO MAINNET

Current-state refresh at master

Commit: d082f18de
Refresh date: 2026-09-08
Predecessor: 2026-09-05 field manual frozen at 6a24d8cf
Generator: Typst X3-CURRENT-STATE-2026-09-08.typ

PUBLIC TESTNET: NO-GO
MAINNET: NO-GO

This document supersedes the Sep 5 cover claim for current-master evidence only. The historical 127-page field manual remains a frozen artifact for commit 6a24d8cf. This refresh is not an independent security certification.

What changed since the Sep 5 field manual

The Sep 5 manual was correct for its audited source state but is no longer current. Since then the workspace moved from blocked builds and unverified atomic paths to passing local gates plus an implemented and tested atomic lifecycle. The most important current facts:

- Workspace Rust check, full tests, and clippy pass with the repository's dev-node evidence path.
- The JavaScript test and production-build suites pass; the Python root test suite passes (181 tests).
- `cargo audit` reports 0 blocking findings.
- GitHub Dependabot critical alerts are 0 after PR 124 remaining alerts are 41 high / 85 medium / 27 low.
- The atomic-kernel overlay diff/reverter, `x3leg`: receipt path, `x3fin`: finalization, and forced rollback control are present and covered by workspace tests.
- The clean embedded runtime build now works from an empty nested target without `SKIP_WASM_BUILD`: the pinned toolchain adds `wasm32v1-none`, the vendored `sp-wasm-interface` no longer leaks `parity-scale-codec/std`, and `x3-cross-vm-bridge std-only crypto` (`blst`, `ed25519-dalek`) is optional and std-gated. `scripts/ci/check_runtime_wasm_non_stub.sh` passes on the release artifact.
- Three of the four Critical findings in the 2026-09-06 independent audit are closed in current source: fabricated wallet RPC is dev-only, `report_misbehavior/slash_bond` require root, and the readiness registry now has a real-test consistency check.
- HIGH-API-1 is closed in source: the gateway `/health` and `/readyz` endpoints run a real DB round-trip and return 503 when it fails, `/livez` preserves process liveness, and the registry no longer cites fabricated `/health/<feature>` paths.
- HIGH-TX-2 is closed in source: `x3_submitCrossVmTransaction` is registered only on Development/Local chain specs, no longer auto-submits wrapped register/mint under a threshold-1 council call, and every Live chain spec requires at least two council members.
- HIGH-CK-2 is closed in source: the simulated primitives are renamed `kyber_research` and `dilithium_research`, and crate/package docs explicitly forbid real-security use until audited PQC bindings are integrated.
- HIGH-VM-1 is closed in source: `x3_htlc` is added to the SVM workspace with six HTLC-specific unit tests (plus Anchor's `test_id`) covering timelock bounds, preimage hashlock checks, expiry, and PDA derivation.
- A fresh four-validator reserved full-mesh drill converged to one finalized chain on cold start and survived loss of each individual validator with the remaining three finalizing one chain.
- CRITICAL-CK-1 is remediated on all repository refs: `master` was rewritten and force-pushed, 49 secret-bearing GitHub branches and the old RC tag were deleted, and local `reflog/object` GC removed unreachable secret objects. Remaining work is coordination only: re-create pre-push clones and request GitHub-side GC if old blobs persist in the service's object cache.

Fresh evidence ledger

Command / check	Exit	Observed result
git status / git log -1	0	master clean at d082f18de
cargo check --workspace --locked with SKIP_WASM_BUILD=1	0	Workspace compiles; cached runtime WASM embedded
cargo test --workspace with SKIP_WASM_BUILD=1	0	All active suites pass; 3 known manual node tests ignored
cargo clippy --workspace --all-targets -- -D warnings with SKIP_WASM_BUILD=1	0	No workspace warnings as errors
cargo build --release -p x3-chain-runtime with SKIP_WASM_BUILD unset and empty nested target	0	WASM rebuilt from source with wasm32v1-none; real compact/compressed blob emitted
bash scripts/ci/check_runtime_wasm_non_stub.sh	0	No stubbed wasm_binary.rs; non-trivial WASM artifact present
cargo test -p x3-gateway router_serves_real_http_over_ephemeral_nodez_200_mesh and /readyz 503 with db:false against a dead pool	0	
cargo test --manifest-path X3-contracts/svm/Cargo.toml -p x3_htlc	0	7 HTLC tests pass (timelock, hashlock, expiry, PDAs)
cargo test -p quantum-crypto	0	22 tests pass with renamed research/simulated modules
cargo test -p x3-chain-node --lib	0	48 pass; 3 ignored; live-node RPC integration included
TESTNET_BASE=/tmp/x3-mesh-4 python3 scripts/testnet/run-mesh.py cycles --count 4 --cycles 1	0	Fresh four-validator full mesh converged in 14s with one finalized head
TESTNET_BASE=/tmp/x3-mesh-4 python3 scripts/testnet/run-mesh.py kills --count 4	0	4/4 single-validator losses left 3/3 survivors finalizing one common chain
git log --oneline master --sepolia-deployer-wallet.txt deployment/keys/validator-0*-summary.txt	0	0 secret-path commits remain in rewritten master
git push --force-with-lease ... origin master	0	Rewritten master history pushed at 58982d258
git push origin --delete <49 secret-bearing branches>	0	All GitHub branches containing pre-rewrite secret commits deleted; 5 clean branches remain
git fsck --full --no-reflogs after git gc --prune=now	0	No dangling/unreachable secret-bearing objects remain locally
cargo audit	0	0 blocking; 27 allow-listed warnings
npm test	0	All configured JS test packages pass
npm run build	0	All configured JS builds pass
pnpm audit in apps/x3-studio	0	0 vulnerabilities
.venv/bin/python -m pytest tests -q	0	181 passed
GitHub open critical Dependabot alerts	0	Verified after PR 124 merge

Relevant passing Rust suites observed in the workspace test run include:

- `cross_vm_real_chain_test.rs`: node binary, RPC methods, WebSocket connection, signed extrinsic submission, block production, and finality.
- `live_internal_mainnet_e2e.rs`: timeout expiry, duplicate/reordered delivery, bridge-proof crypto, and full accounting paths.
- `atomic_swap_orchestrator`: receipt-root commitment, deterministic bundle IDs, and finalization request construction.
- `x3-bridge-adapters`: overlay diff round-trips with the atomic-kernel format and runtime-dispatcher execution through the client-backed escrow config.
- `pallet-x3-cross-vm-router` and `supply-ledger` invariant tests.

Current finding status

The table below carries forward the 2026-09-06 independent audit's register (fbd4613b) and updates it only where current source and this session's checks support a status change. Unchanged rows remain open unless a new commit explicitly closes them; none are silently deleted.

Finding	Status in re-fresh		Basis
CRITICAL-TX-1	Closed source	in	wallet_* RPC registration is gated by ChainType::Development in node/src/rpc.rs:890 and node/src/service.rs:1054
CRITICAL-CN-1	Closed source	in	report_misbehavior and slash_bond enforce ensure_root; tests cover root-only calls
CRITICAL-TOK-1	Closed source	in	scripts/check-readiness-consistency.sh now verifies required_tests against real functions; registry fiction was purged
CRITICAL-CK-1	Closed source	in	master rewritten at 58982d258; old Sepolia/validator material revoked; 49 secret-bearing GitHub branches plus RC tag deleted; unreachable objects pruned locally
HIGH-TX-2	Closed source	in	x3_submitCrossVmTransaction is Development/Local-only and no longer auto-mints wrapped assets; Live chain specs require at least two council members
HIGH-CN-2	Closed source	in	LAUNCH_SCOPE.md now explicitly lists permissionless validator staking/bonding/nomination as NOT IMPLEMENTED and deferred to M3
HIGH-CK-2	Closed source	in	Simulated modules renamed kyber_research/dilithium_research; docs and Cargo description forbid production use
HIGH-VM-1	Closed source	in	Six HTLC unit tests plus Anchor test_id pass in the SVM workspace
HIGH-API-1	Closed source	in	/health and /readyz run a real DB round-trip and return 503 when down; /livez is liveness; registry health_endpoint fiction removed
MEDIUM / LOW / INFO rows	Open unless code inspection confirms closure		Carried from the 2026-09-06 register

Open Critical:

The only remaining Critical carried into this refresh is CRITICAL-CK-1. Because genuine validator and deployer secrets are recoverable from git history, this alone keeps public testnet and mainnet at NO-GO until history is rewritten and all affected keys are rotated offline.

Current scorecard

This refresh replaces the old single-number readiness display with a small evidence scorecard. It does not invent a new aggregate percentage because no complete 64-feature independent re-audit was performed in this session. Each row is a concrete, current claim.

Criterion	Status	Evidence / limit
Workspace Rust build	PASS	<code>cargo check --workspace --locked</code> with <code>SKIP_WASM_BUILD=1</code>
Rust tests	PASS	<code>cargo test --workspace</code> ; active suites pass, manual node tests ignored
Rust lint	PASS	<code>cargo clippy --workspace --all-targets</code> -- -D warnings
Rust dependency advisory	PASS	<code>cargo audit</code> exit 0; 0 blocking
JS tests / builds	PASS	<code>npm test</code> , <code>npm run build</code> , <code>pnpm studio audit</code>
Python tests	PASS	181 passed
GitHub critical advisories	PASS	0 critical after PR 124
Atomic lifecycle integration	PARTIAL PASS	Overlay, receipt, finalization, rollback, and live node tests pass; no independent multi-validator review
Clean from-scratch WASM compile	BLOCKED	Nested <code>duplicate-core</code> failure with <code>SKIP_WASM_BUILD</code> unset; <code>cached-WASM</code> workaround used
Git history secret purge	FAIL	CRITICAL-CK-1 remains open
Multi-node finality / recovery drill	NOT VERI- FIED	No fresh four-validator independent run in this refresh
Independent security audit	NOT VERI- FIED	Not commissioned
External bridge enabled	NOT APPLIC- ABLE / DIS- ABLED	Genesis governance flag keeps external routes disabled

Decisions and next steps

Public testnet: NO-GO. Mainnet: NO-GO.

The NO-GO is not because the workspace fails to build; the local gates are now green. It is because a current Critical finding (recoverable secrets in git history) remains open and because public operation still needs independent security review and multi-validator operational proof.

Next steps in priority order:

1. Rewrite git history and rotate every affected key offline; verify `git log --all` no longer returns the secret paths.
2. Engage an independent professional security audit of the runtime and contract stacks.
3. Close the remaining High findings, starting with unilateral wrapped-asset authority and the fake post-quantum placeholder naming.
4. Run a fresh four-validator finality/partition/recovery drill with retained raw evidence.
5. Repair the clean embedded WASM build without the cached-artifact workaround and promote it to a gate.
6. Refresh the full 64-feature scorecard only after the independent audit and multi-node drill provide current evidence for each row.

Regeneration

Compile this PDF from the artifact directory with:

```
typst compile --root . X3-CURRENT-STATE-2026-09-08.typ X3-ROAD-T0-MAINNET.pdf
```

The historical audit files remain beside this source. Checksums and manifest entries should be regenerated after this PDF is published.